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Figure 1: Rome from Pincian Hill, Italy. The spire of All Saints' Anglican Church rises above the historic cityscape, with St. Peter's Basilica visible on the horizon. Photograph by Yoni Rodriguez, November 2025.

SHM 102: Occupational Health

Public Health History

Lecture 02 · Fall 2026

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From Rome to Sicily

Our journey today begins in Rome, where the opening lecture photograph was taken from Pincian Hill in November 2025.

From Rome, we travel south through Italy to Messina, Sicily, a port city at the northeastern tip of the island and the starting point for our historical case.

Explore the map below. *Where is Messina in relation to Rome and the Mediterranean Sea?*

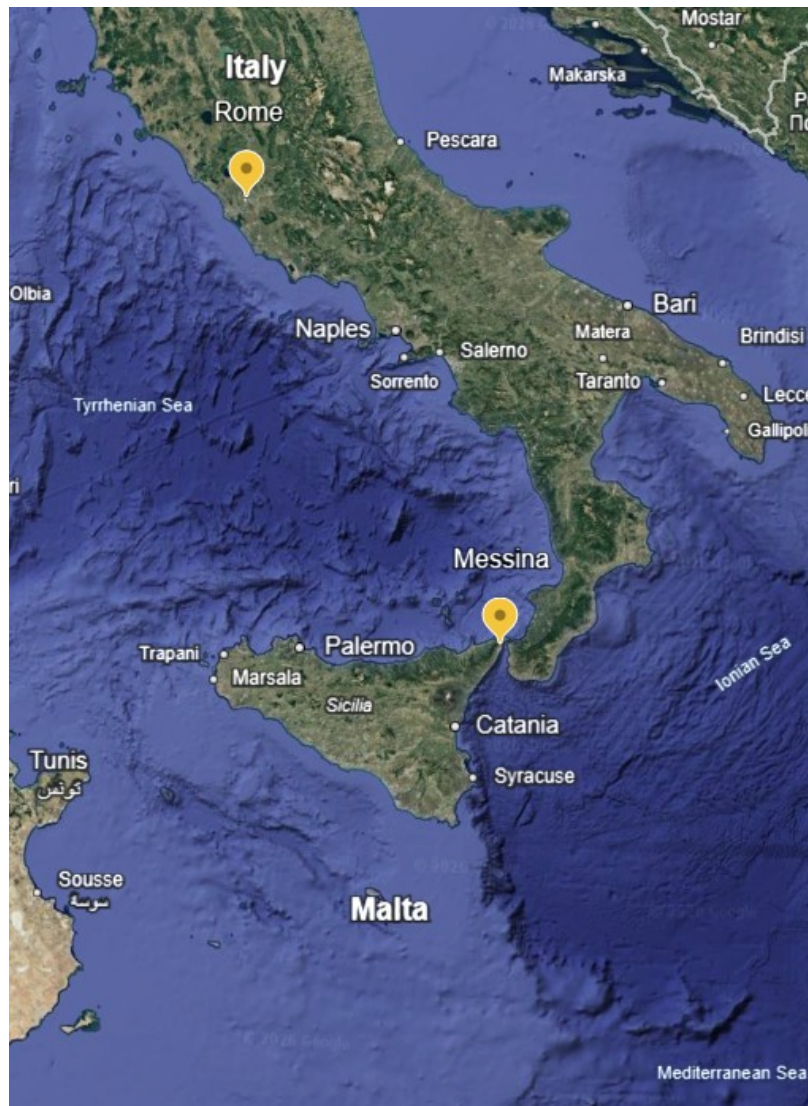


Figure 2: Google Earth map showing Rome and Messina in relation to Italy, Sicily, and the Mediterranean Sea.

Want to explore further?

[Explore Rome and Messina in Google Earth](#)

Interactive Google Earth project created by Yoni Rodriguez.

Travel back to 1347

October 1347 · Messina, Sicily



Figure 3: Historic painting of a Mediterranean harbor with sailing ships, merchants, travelers, animals, and cargo.

*Hendrik van Minderhout, **Italianate harbour scene with the monument of Ferdinand I de' Medici at Leghorn**, 1670. National Maritime Museum, Greenwich. The painting depicts a later Mediterranean harbor scene and is used here to help visualize the maritime setting of our 1347 case.*

Something is wrong.

Merchant ships have arrived from the **Black Sea**.

Most of the sailors aboard are **dead**.

Many of those still alive are **gravely ill**.

You are the Port Health Officer, What do you do?

What would you want to know?

Poll Everywhere

Before deciding what to do, *what information would help you understand what is happening?*

What questions would you ask?

Submit one question through Poll Everywhere.



What are we trying to figure out?

As a group, your questions may begin to fall into a few categories:

- **Who** is sick?
- **Where** did they come from?
- **When** did they become sick?
- **What symptoms** are they experiencing?
- **What** did they have in common?
- **What** might they have been exposed to?
- **Who** else may have been exposed?
- Is the illness spreading?

We still do not know what is causing the illness.

You have to make a decision.

You still do not know what is causing the illness.

Poll Everywhere

What should you do with the arriving ships?

Choose the response you think is most appropriate.

1. Allow passengers and crew to disembark normally
2. Allow only people who appear healthy to disembark
3. Isolate the sick but allow everyone else ashore
4. Keep the ships and everyone aboard separated from the population
5. I need more information before deciding



How certain do you need to be before taking action?

You know that people are **severely ill**.

You know that many have **died**.

You know that the ships have traveled from somewhere else.

But you **do not know the cause**.

Do you wait until you know more? Or act now to prevent additional illness?

Prevention before understanding

One response seems intuitive:

Keep sick people away from healthy people.

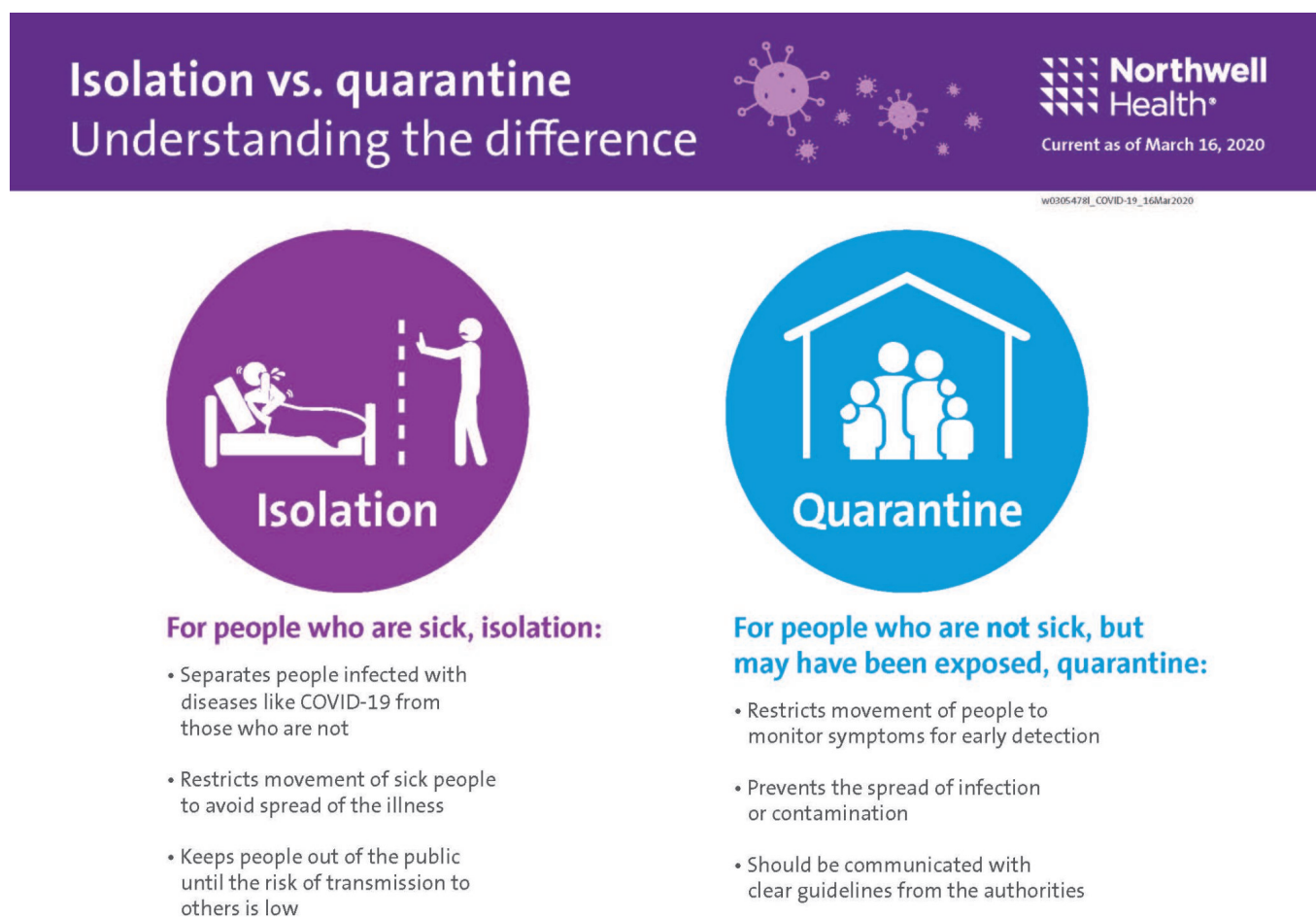
You may not know **what** is causing the illness.

You may not know **how** it spreads.

But you can still try to prevent additional illness.

Isolation or quarantine?

These ideas are related, but they are not the same.



Sources: CDC, What To Do If You Are Sick and WHO, Considerations for quarantine of individuals in the context of containment for coronavirus disease (COVID-19)

Figure 4: Northwell Health infographic comparing isolation and quarantine.

Northwell Health, CDC and WHO guidance. [View source](#)

Which one did you recommend for the ships?

From separation to quarantine

As epidemics continued to threaten Mediterranean cities, communities began developing more formal ways to separate potentially infected travelers, ships, and goods.

Another Mediterranean port

In **1377**, the Adriatic port of **Ragusa** (modern-day) **Dubrovnik, Croatia** introduced one of the earliest formal systems for separating travelers arriving from areas affected by disease.



Figure 5: Google Earth map showing Rome, Messina, and Dubrovnik in relation to the Mediterranean and Adriatic Seas.

Want to explore further?

[Explore the Mediterranean locations in Google Earth](#)

Interactive Google Earth project created by Yoni Rodriguez.

How quarantine got its name

Travelers arriving in Ragusa from areas affected by disease were required to remain separated before entering the city.

The initial period was **30 days**.

Later practices extended the period to **40 days**.

Italian: *quaranta giorni*

“forty days”

→ quarantine

But there is a problem.

You decided to keep sick sailors away from the population.

What else is aboard a merchant ship?

People are not the only things that travel.

Merchant ships move:

- people
- animals
- food
- water
- cargo
- clothing and personal belongings

Could something else aboard the ship be contributing to the illness?

How did we get here?

For most of human history, people lived in relatively small and mobile populations.

Around **12,000 years ago**, the development of agriculture began changing how many humans lived.



Figure 6: Rock paintings depicting Neolithic farming and interactions between people and domesticated animals in Tassili de Maghinet, Libya.

Source

Humans were changing their relationship with animals and the environment.

Agriculture changed exposure

Living in permanent settlements brought people into closer and more frequent contact with:

- other people
- domesticated animals
- stored food
- animal and human waste
- water sources
- rodents and other wildlife

At the same time, populations became larger and increasingly connected.

New ways of living created new opportunities for exposure and disease.

Health exists within a system

The relationships we have been describing are still important today.

Human health does not exist independently from animals or the environment.

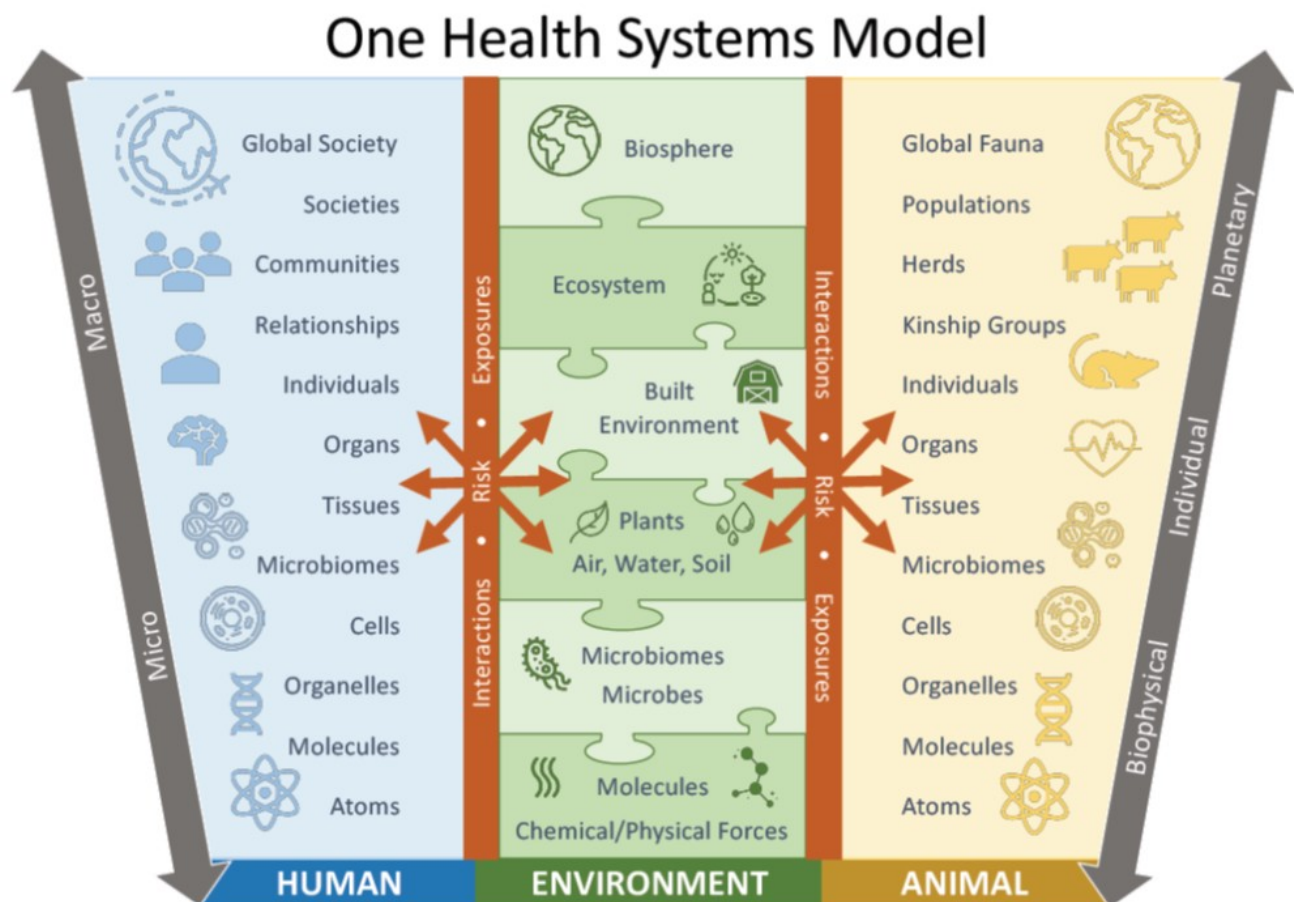


Figure 7: One Health Systems Model showing interactions among human, environmental, and animal systems across biological and societal scales.

Source: ENVH 501 Foundations of Environmental Health, University of Washington, “EOH Approach 1,” 2020.

Where would our merchant ship fit into this system?

One Health

The health of humans, animals, and the environment are interconnected.

Our merchant ship brings all three together:

Human

Sailors, travelers, port workers, and surrounding communities.

Animal

Domesticated animals and other animals moving with or around people.

Environment

Ships, ports, food, water, cargo, waste, and the built environment.

Changes in one part of the system can create risks in another.

Trade connected populations

Ports such as Messina were part of larger transportation networks connecting communities across the Mediterranean and beyond.

Ships did more than transport people. They connected cities, markets, animals, food, goods, and environments.

Transportation networks can also become pathways through which health threats move.

What have we been doing?

Throughout this case, we asked:

- **Who** is getting sick?
- **Where** is it happening?
- **When** did illness begin?
- **What** might people have been exposed to?
- **Who else** might be at risk?
- **What can we do** to prevent additional illness?

This is public health.

What is public health?

Public health focuses on the health of **populations** rather than only the treatment of individual patients.

It asks how disease, injury, and other health threats can be:

- recognized
- investigated
- prevented
- controlled

Public health asks: How can we prevent harm before more people become sick?

673 years later...

March 2020

A cruise ship approaches California after days awaiting clearance to come ashore.

There are confirmed cases of a newly emerging infectious disease aboard and potentially more.

Cruise Ship, Floating Symbol of America's Fear of Coronavirus, Docks in Oakland

After days awaiting clearance to come ashore, the Grand Princess arrived in the San Francisco Bay with 21 confirmed coronavirus cases on board, and possibly more.



Share full article



34



Passengers will be whisked to quarantine on military bases, including nearby Travis Air Force Base. *Jon Wozniak/The New York Times*

Figure 8: New York Times coverage of the Grand Princess cruise ship arriving in San Francisco Bay during the COVID-19 outbreak in March 2020.

The New York Times, March 9, 2020. [Read the article](#)

Public health officials must decide:

- Who can leave the ship?

- What were they exposed to?
 - Who may have been exposed?
 - Where should passengers go?
 - Who should be tested?
 - How can additional illness be prevented?
-

And then it happened again.

Spring 2026

An expedition cruise ship is linked to a cluster of severe illness.

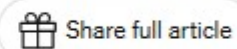
People have died.

Others may have been exposed.

Travelers from multiple countries have been aboard.

3 Dead of Suspected Hantavirus Infections on Cruise Ship, W.H.O. Says

One case of hantavirus infection has been confirmed in a laboratory, the organization said on Sunday. There are five additional suspected cases.



The cruise ship MV Hondius off Cape Verde on Sunday. Officials identified the ship as the one with passengers infected by hantavirus. Agence France-Presse — Getty Images

Figure 9: New York Times coverage of suspected hantavirus infections associated with the MV Hondius expedition cruise ship in May 2026.

The New York Times, May 3, 2026. [Read the article](#)

Public health officials again have to ask:

- Who is sick?
 - Who may have been exposed?
 - Where have they traveled?
 - Who else needs to be contacted?
 - Is the illness spreading?
 - What should be done while the outbreak is investigated?
-

Our ability to investigate disease has changed dramatically.

The need to make decisions under uncertainty has not.

2026: Hantavirus | 2020: SARS-CoV-2 | 1347: _ _ _

What arrived in Messina, Sicily in 1347?

What tools, methods, and technologies can help us determine this?

Before Monday...

Think about the people living in 1347. Could they:

- observe illness?
- observe the cause?
- separate sick people?
- restrict the spread?
- how effective their attempts were?

Key Terms

Term	What you should know
Public health	Protecting and improving the health of populations through prevention and organized action
Disease	A condition that disrupts normal functioning and produces adverse health effects
Epidemic	Disease occurrence in a population above what is normally expected
Isolation	Separating people who are ill from people who are not ill

Term	What you should know
Quarantine	Restricting the movement of people who may have been exposed to disease
Imported case or outbreak	A case or outbreak introduced into a population from another geographic location
Transportation network	A connected system through which people, animals, and goods move between places
One Health	An approach recognizing that the health of humans, animals, and the environment are interconnected
How did the Agricultural Revolution change exposure?	Permanent settlements increased population density and brought humans into closer contact with domesticated animals, stored food, waste, water sources, rodents, and other wildlife, creating new opportunities for exposure and disease transmission

Make your flashcards

Copy → Paste → Study

Copy the terms below to make your own **Anki** or **Quizlet** flashcards.

Anki: Save as a .txt file → Import → Select **Semicolon** as the field separator.

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SHM 102: Occupational Health

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